

INTEGRALI DOPPI SU DOMINI NORMALI

$$1) f(x,y) = \frac{\sin(y^2)}{y} \quad I = \iint_{\Omega} f(x,y) dx dy$$

$$\Omega = \{(x,y) \in \mathbb{R}^2 : 0 \leq y \leq \sqrt{\pi}, 0 \leq x \leq y^2\}$$

$$\Omega \text{ è } y\text{-NORM} \quad y \in [c,d] = [0, \sqrt{\pi}]$$

$$g(y) \leq x \leq h(y)$$

$$g(y) = 0 \quad h(y) = y^2$$

$$g, h: [c,d] \rightarrow \mathbb{R}$$

$$g, h: [0, \sqrt{\pi}] \rightarrow \mathbb{R}$$

$$I = \int_0^{\sqrt{\pi}} \int_0^{y^2} \frac{\sin y^2}{y} dx dy =$$

$$= \int_0^{\sqrt{\pi}} \frac{\sin y^2}{y} \cdot y^2 dy = \int_0^{\sqrt{\pi}} y \cdot \sin y^2 dy =$$

$$-\frac{1}{2} \cos(y^2) \text{ È UNA PRIMITIVA DI } y \cdot \sin y^2$$

$$= -\frac{1}{2} \cos y^2 \Big|_0^{\sqrt{\pi}} = 1$$

2) $f(x,y) = x \cdot y$

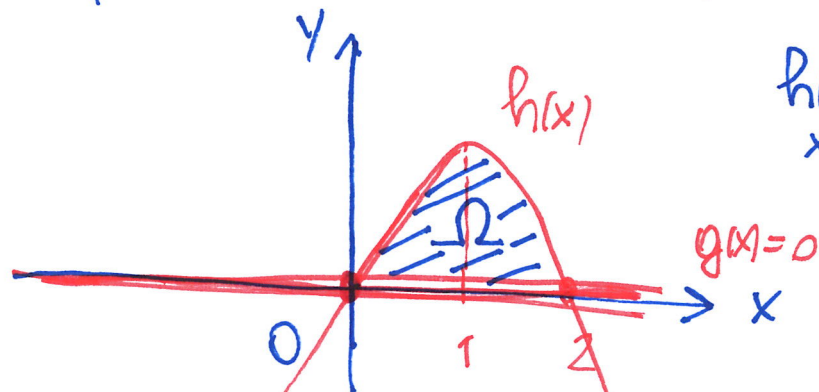
$\Omega = \{(x,y) \in \mathbb{R}^2 : 0 \leq y \leq 2x - x^2\}$ $I = \iint_{\Omega} xy \, dx \, dy$

Ω È X-NORMALE

$0 = g(x) \leq y \leq h(x) = 2x - x^2 \quad x \in [a,b]$
 $g, h: [a,b] \rightarrow \mathbb{R}$

RAPPRESENTARE Ω SUL PIANO $\rightarrow$ DISEGNARE LA g E LA h
 COME SE IL DOMINIO

FOI E TUTTO $\mathbb{R}$



$[a,b] = [0,2]$

$I = \int_0^2 \int_0^{2x-x^2} xy \, dy \, dx =$

$= \int_0^2 x \left[\frac{1}{2} y^2 \right]_0^{2x-x^2} dx = \frac{1}{2} \int_0^2 x (2x-x^2)^2 dx =$

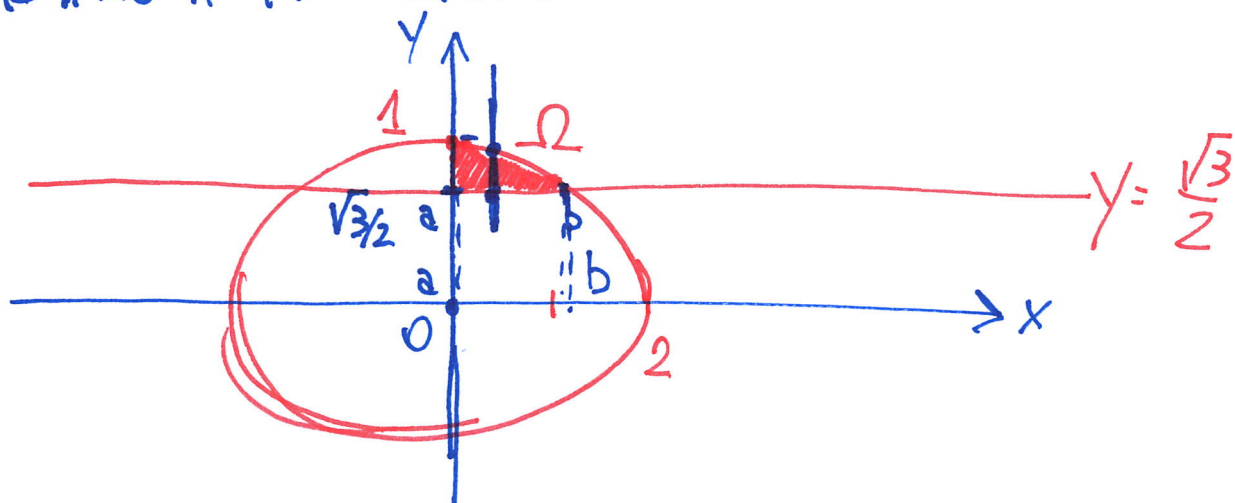
$= \frac{1}{2} \left(\int_0^2 (x^5 + 4x^3 - 4x^4) dx \right) = \dots = \frac{1}{2} \left(16 + \frac{64}{6} - \frac{128}{5} \right)$

$$3) \quad I = \iint_{\Omega} 2x \, dx \, dy$$

(3)

$$\Omega = \left\{ (x,y) \in \mathbb{R}^2 : x \geq 0, y \geq \frac{\sqrt{3}}{2}, \frac{x^2}{4} + y^2 \leq 1 \right\}$$

PROVIAMO A DISEGNARLO!!



Ω È X-NORM

$$x \in [a,b] = [0,1]$$

$$g(x) \leq y \leq h(x) \quad g, h: [a,b] \rightarrow \mathbb{R}$$

$$g(x) = \frac{\sqrt{3}}{2}$$

$$y(x) = \sqrt{1 - \frac{x^2}{4}} = h(x)$$

$$y^2 = 1 - \frac{x^2}{4}$$

$$a=0 \quad b=?$$

$$\begin{cases} y = \frac{\sqrt{3}}{2} \\ \frac{x^2}{4} + y^2 = 1 \end{cases}$$

$$\frac{x^2}{4} + \frac{3}{4} = 1 \quad \frac{x^2}{4} = \frac{1}{4} \quad x^2 = 1$$

$$x = \pm 1 = b$$

$$I =$$

$$\int_0^1 \int_{\frac{\sqrt{3}}{2}}^{\sqrt{1 - \frac{x^2}{4}}} 2x \, dy \, dx$$

(4)

$$= \int_0^1 2x \left(\sqrt{1 - \frac{x^2}{4}} - \frac{\sqrt{3}}{2} \right) dx =$$

$$= \int_0^1 2x \sqrt{1 - \frac{x^2}{4}} dx - \underbrace{\sqrt{3}}_{\frac{\sqrt{3}}{2}} \int_0^1 x dx$$

$$\int g^\alpha(x) \cdot g'(x) dx = \frac{g^{\alpha+1}(x)}{\alpha+1} \quad \alpha \neq -1$$

$$g(x) = 1 - \frac{x^2}{4} \quad \alpha = \frac{1}{2}$$

$$g'(x) = -\frac{1}{2}x$$

$$= -4 \left(\int_0^1 -\frac{1}{2}x \sqrt{1 - \frac{x^2}{4}} dx \right) - \frac{\sqrt{3}}{2}$$

$$= -4 \left[\frac{2}{3} \left(1 - \frac{x^2}{4} \right)^{3/2} \right]_0^1 = -\frac{8}{3} \left[\left(\frac{3}{4} \right)^{3/2} - 1 \right] - \frac{\sqrt{3}}{2}$$

$$4) I = \iint_E \sin(y^3) dx dy$$

$$E = \{(x,y) \in \mathbb{R}^2 : x \in [0,1], \sqrt{x} \leq y \leq 1\}$$

$$E \hat{=} x\text{-MORM} \quad x \in [a,b] = [0,1]$$

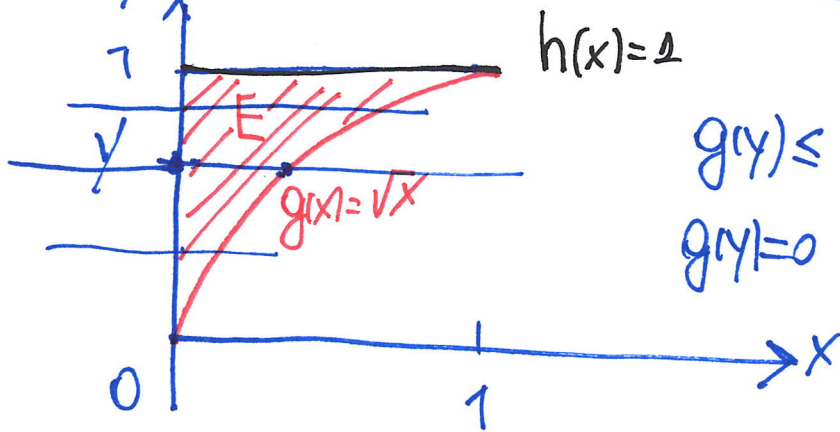
$$\sqrt{x} = g(x) \leq y \leq h(x) = 1$$

(5)

$$I = \int_0^1 \int_{\sqrt{x}}^1 \sin(y^3) dy dx$$

$$y \in [c, d] = [0, 1]$$

E y-NORMALE



$$g(y) \leq x \leq h(y) = y^2$$

$$g(y) = 0$$

$$y = \sqrt{x} \quad x = y^2$$

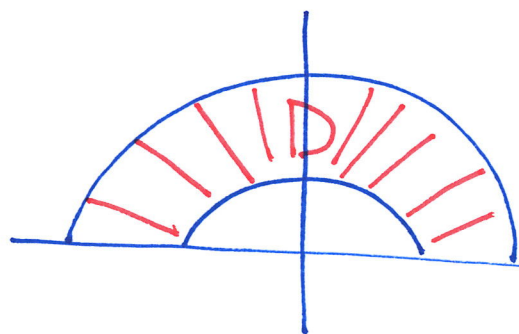
$$I = \int_0^1 \int_0^{y^2} \sin(y^3) dx dy$$

$$= \int_0^1 y^2 \sin(y^3) dy = \frac{1}{3} \int_0^1 3y^2 \sin(y^3) dy =$$

$$= \frac{1}{3} [-\cos(y^3)]_0^1 = \frac{1}{3} (1 - \cos(1)). \quad \square$$

FORMULA DEL CAMBIAMENTO DI VARIABILI

6



$$D \rightarrow D'$$

PIÙ SEMPLICE DI D
→ OBIETTIVO MASSIMO
D' È UN RETTANGOLO!!

↓ PER FUNZIONI A 1 VARIABILE SIGNIFICA
INTEGRAZIONE PER SOSTITUZIONE

$$\int_a^b f(x) dx$$

$$\phi: [c, d] \rightarrow [a, b]$$

$$\begin{cases} \phi \in C^1([c, d]) \\ \phi \text{ INVERTIBILE} \end{cases}$$

UNA COND. SUFF. CHE VI GARANTISCE
è che $\phi'(t) \neq 0 \quad \forall t \in]c, d[$
 > 0

$$x = \phi(t)$$

$$\int_a^b f(x) dx = \int_c^d f(\phi(t)) \cdot \phi'(t) dt$$

(7)

$$I = \iint_D f(x,y) dx dy = ?$$

$$D \quad T: D' \rightarrow D$$

$$T(u,v) = (x,y)$$

$$D' \subseteq \mathbb{R}^2$$

$$\begin{cases} x = g(u,v) \\ y = h(u,v) \end{cases}$$

$$\begin{cases} g, h \in C^1(D') \end{cases}$$

$$\begin{cases} \text{~~g~~ } T \text{ INVERTIBILE} \end{cases}$$

UNA COND. SUFF. $\uparrow$ PER GARANTIRE LA 2^a MPI È CHE....

$$J_T(u,v) = \begin{pmatrix} \frac{\partial g}{\partial u} & \frac{\partial g}{\partial v} \\ \frac{\partial h}{\partial u} & \frac{\partial h}{\partial v} \end{pmatrix}$$

MATRICE
JACOBIANA
di T

$$\dots |J_T(u,v)| = \det J_T(u,v) > 0$$

A

ALMENO IN UN SOTTOINSIEME di D' TALE CHE

~~D'~~ $D' \setminus A$ È di MISURA NULLA *

E

$T(D' \setminus A)$ È di MISURA NULLA

* ESEMPI

UN PUNTO
UN SEGMENTO
UN ARCO di CURVA

→ di MISURA ZERO

$$\iint_D f(x,y) dx dy = \iint_{D'} f(g(u,v), h(u,v)) |J_T(u,v)| du dv$$

FORMULA DALLE COORDINATE CARTESIANE ALLE POLARI



CASO CLASSICO

$$\begin{cases} x = \rho \cos \theta = g(\rho, \theta) \\ y = \rho \sin \theta = h(\rho, \theta) \end{cases} \quad \rho = \sqrt{x^2 + y^2} \quad \begin{matrix} \rho_1 \leq \rho \leq \rho_2 \\ \theta_1 \leq \theta \leq \theta_2 \end{matrix}$$

$T \in C^1$

$$\iint_D f(x, y) dx dy = \iint_{D'} f(\rho \cos \theta, \rho \sin \theta) \rho \, d\rho \, d\theta$$

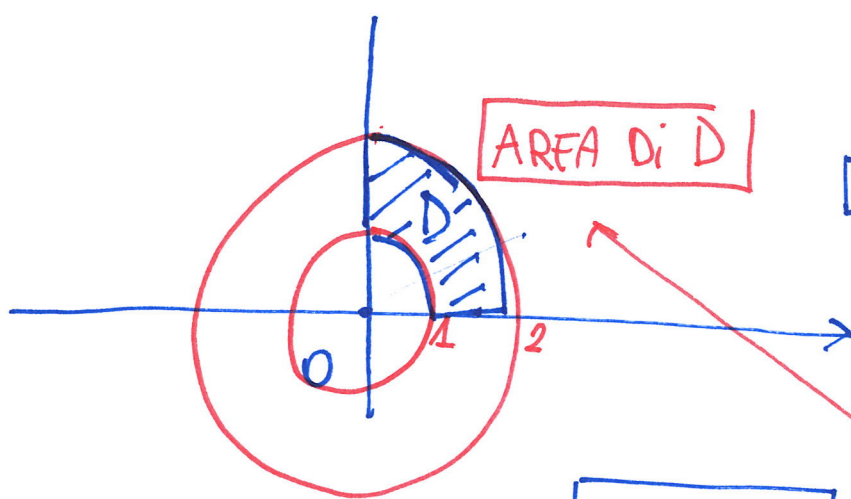
$$J_T(\rho, \theta) = \begin{pmatrix} \cos \theta & -\rho \sin \theta \\ \sin \theta & \rho \cos \theta \end{pmatrix} \quad \downarrow$$

$$|J_T(\rho, \theta)| = \rho \cos^2 \theta + \rho \sin^2 \theta = \rho > 0$$

ESEMPIO 1

$$I = \iint_D dx dy \quad f(x, y) = 1$$

$$D = \{(x, y) \in \mathbb{R}^2 : x \geq 0, y \geq 0, 1 \leq x^2 + y^2 \leq 4\}$$

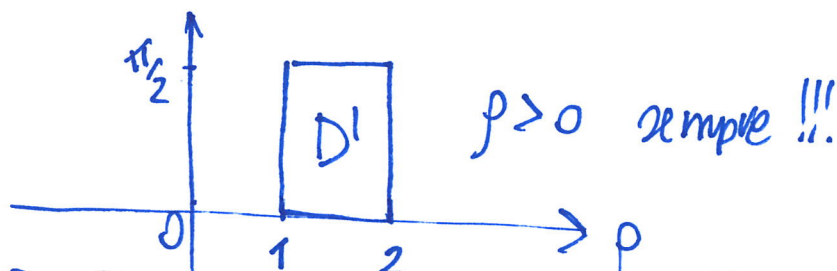


AREA Di D

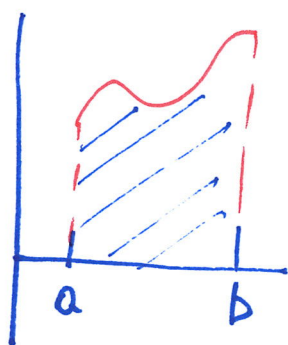
$$D' = \{(p, \theta) : 1 \leq p \leq 2, 0 \leq \theta \leq \pi/2\}$$

$$\begin{cases} x = p \cos \theta \\ y = p \sin \theta \end{cases}$$

$$\begin{aligned} 1 &\leq p \leq 2 \\ 0 &\leq \theta \leq \pi/2 \end{aligned}$$



$$\begin{aligned} I = \iint_D dx dy &= \int_1^2 \int_0^{\pi/2} p \, dp \, d\theta = \left(\int_1^2 p \, dp \right) \cdot \left(\int_0^{\pi/2} d\theta \right) = \\ &= \frac{1}{2} p^2 \Big|_1^2 \cdot \frac{\pi}{2} = \frac{3}{4} \pi. \quad \square \end{aligned}$$



$$\int_a^b f(x) dx = \text{AREA TRAPEZIOIDE}$$

